

## **Windows Security Log Analysis Using Log Parser Studio**

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## 1. Objective

The objective of this lab was to analyze Windows Security Logs using Log Parser Studio and perform event-based investigations through structured queries. Unlike manual Event Viewer analysis, Log Parser Studio enables rapid filtering, aggregation, and investigation of large volumes of security events.

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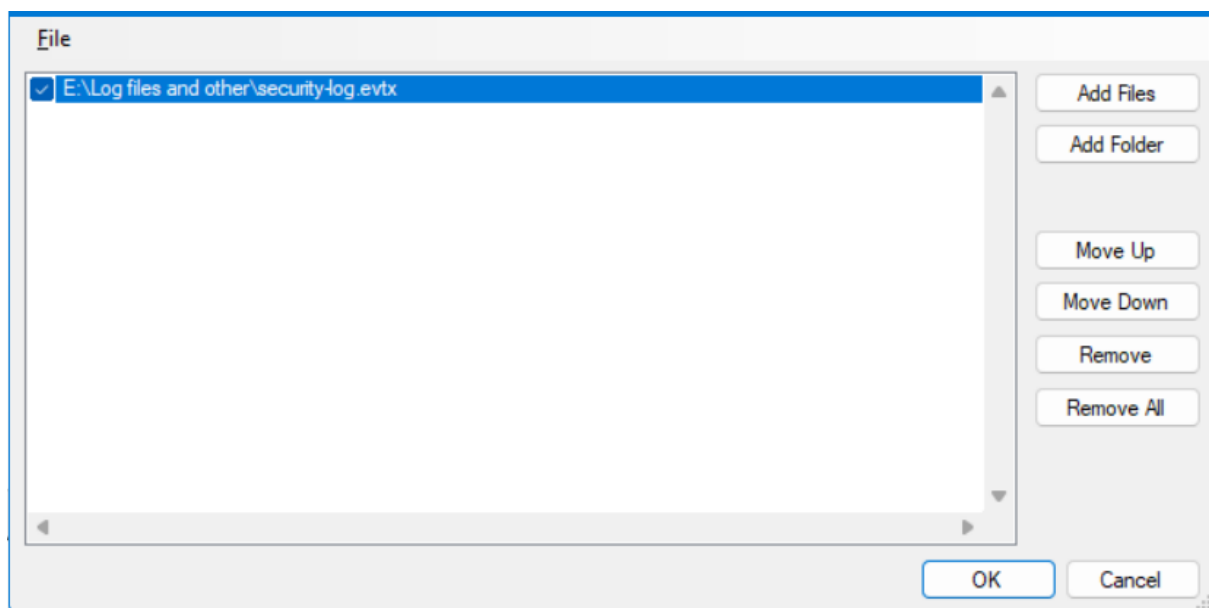
## 2. Lab Environment

|                    |                               |
|--------------------|-------------------------------|
| Operating System   | Windows 11                    |
| Log Source         | Windows Security Log (.evtx)  |
| Analysis Tool      | Log Parser Studio             |
| Investigation Type | Authentication Event Analysis |

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## 3. Loading Security Logs into Log Parser Studio

The exported Windows Security Log was imported into Log Parser Studio for analysis. This allowed event records to be queried using SQL-like syntax rather than manually browsing thousands of entries.



## Observation

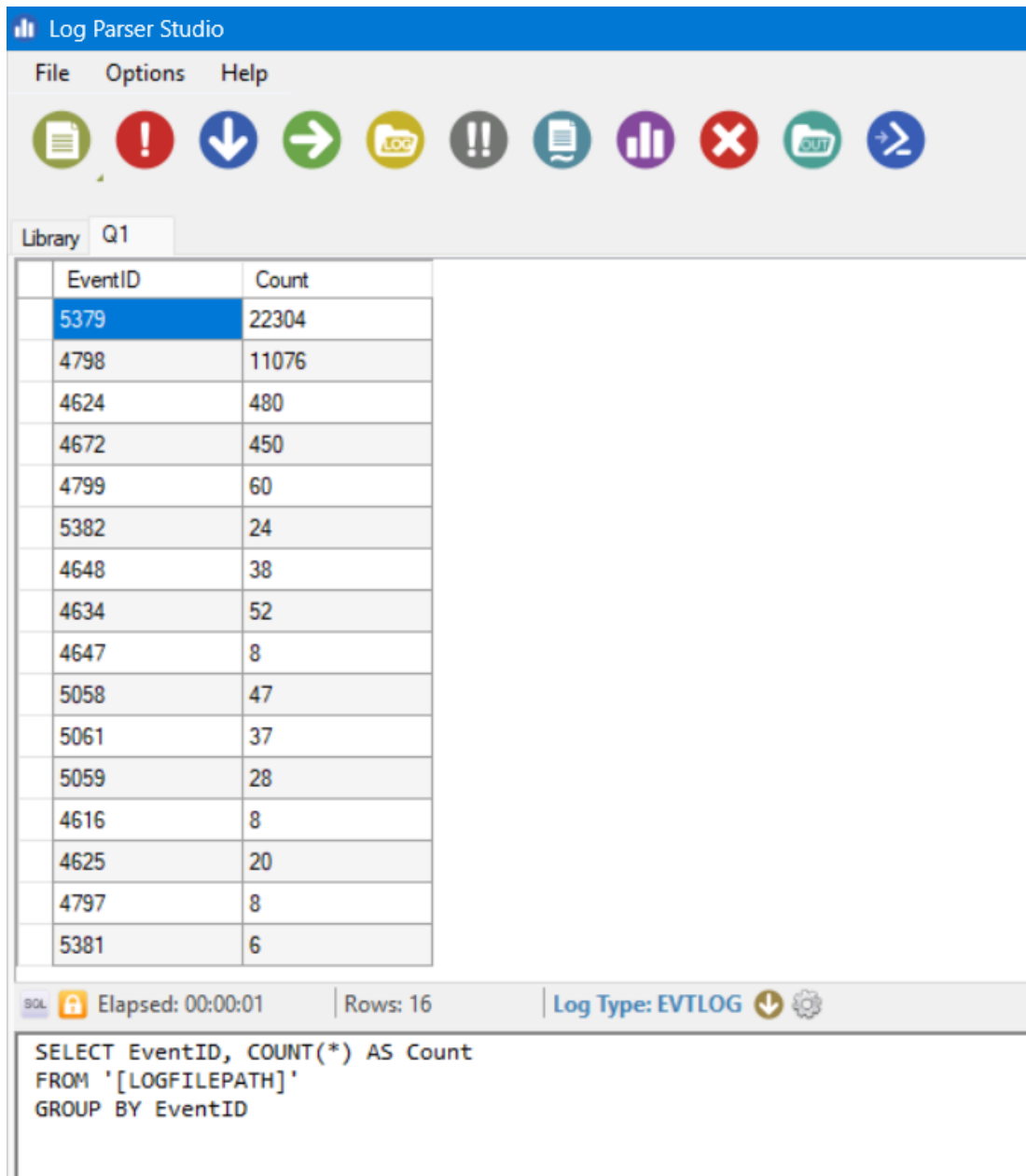
The Security log file was successfully loaded and made available for querying. Log Parser Studio recognized the EVTX source and allowed event data to be analyzed through custom queries.

#### 4. Event Frequency Analysis

To understand the overall distribution of security events, an aggregation query was executed to count occurrences of each Event ID.

##### Query

```
SELECT EventID, COUNT(*) AS Count
FROM '[LOGFILEPATH]'
GROUP BY EventID
```



| EventID | Count |
|---------|-------|
| 5379    | 22304 |
| 4798    | 11076 |
| 4624    | 480   |
| 4672    | 450   |
| 4799    | 60    |
| 5382    | 24    |
| 4648    | 38    |
| 4634    | 52    |
| 4647    | 8     |
| 5058    | 47    |
| 5061    | 37    |
| 5059    | 28    |
| 4616    | 8     |
| 4625    | 20    |
| 4797    | 8     |
| 5381    | 6     |

##### Observation

The query returned a frequency table showing the most common security events recorded within the log.

Several Event IDs appeared repeatedly, indicating routine operating system activity.

## **Analysis**

Reviewing event frequency helps analysts identify:

- Common authentication events
- Credential-related activity
- Privileged account usage
- Unusual spikes in event generation

## **Key Finding**

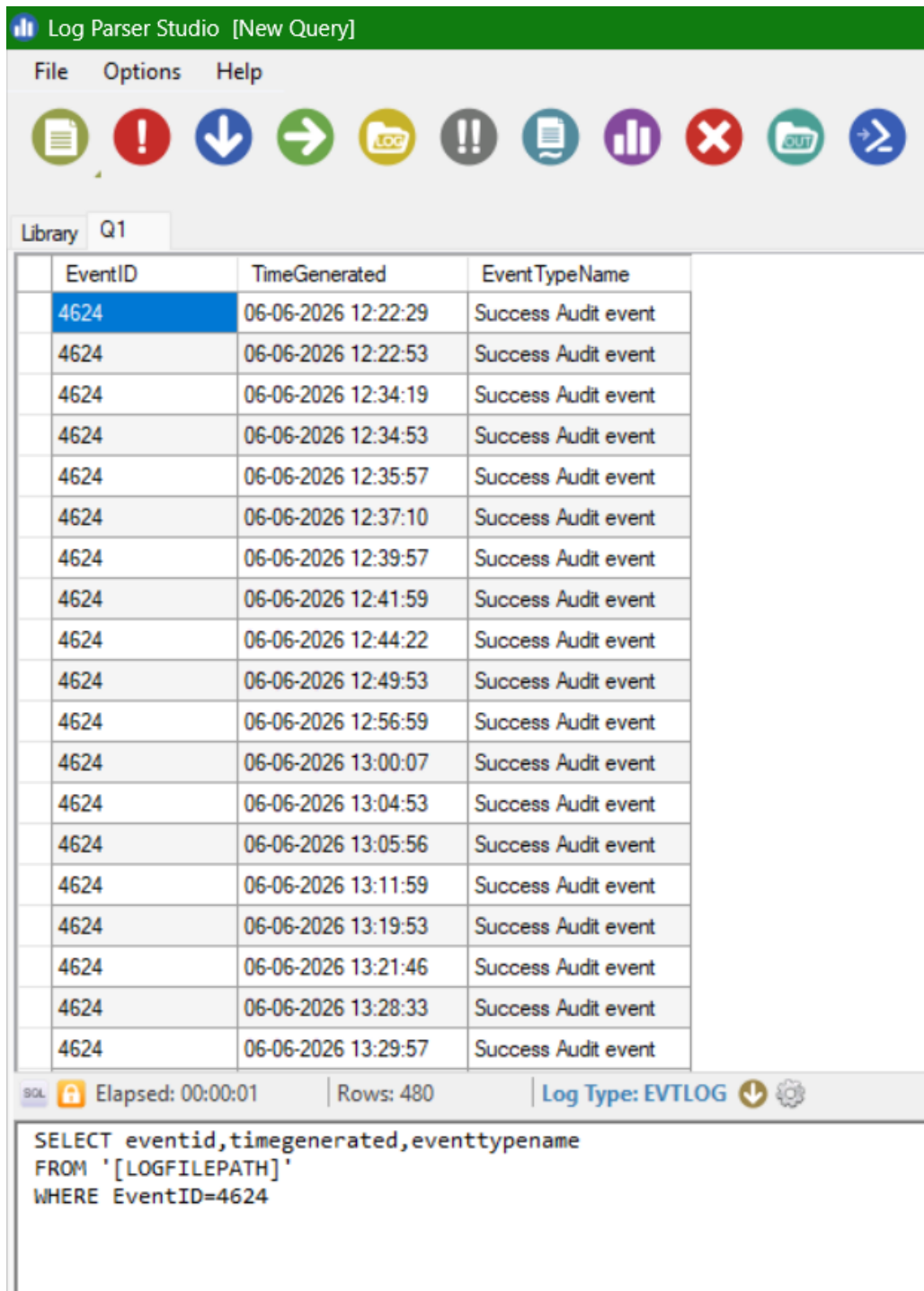
Event ID 5379 appeared most frequently within the dataset, indicating significant Credential Manager activity.

## 5. Successful Authentication Analysis (Event ID 4624)

Event ID 4624 records successful logon activity and is one of the most important authentication events within Windows.

### Query

```
SELECT EventID, TimeGenerated, EventTypeName
FROM '[LOGFILEPATH]'
WHERE EventID = 4624
```



The screenshot shows the Log Parser Studio interface. The title bar reads "Log Parser Studio [New Query]". The menu bar includes "File", "Options", and "Help". The toolbar contains icons for file operations, error handling, and execution. The "Library" tab is active, showing a table with three columns: "EventID", "TimeGenerated", and "EventTypeName". The table contains 20 rows of data, all for EventID 4624, representing successful audit events. The status bar at the bottom indicates "Elapsed: 00:00:01", "Rows: 480", and "Log Type: EVTLOG". The SQL query is displayed in the bottom pane.

| EventID | TimeGenerated       | EventTypeName       |
|---------|---------------------|---------------------|
| 4624    | 06-06-2026 12:22:29 | Success Audit event |
| 4624    | 06-06-2026 12:22:53 | Success Audit event |
| 4624    | 06-06-2026 12:34:19 | Success Audit event |
| 4624    | 06-06-2026 12:34:53 | Success Audit event |
| 4624    | 06-06-2026 12:35:57 | Success Audit event |
| 4624    | 06-06-2026 12:37:10 | Success Audit event |
| 4624    | 06-06-2026 12:39:57 | Success Audit event |
| 4624    | 06-06-2026 12:41:59 | Success Audit event |
| 4624    | 06-06-2026 12:44:22 | Success Audit event |
| 4624    | 06-06-2026 12:49:53 | Success Audit event |
| 4624    | 06-06-2026 12:56:59 | Success Audit event |
| 4624    | 06-06-2026 13:00:07 | Success Audit event |
| 4624    | 06-06-2026 13:04:53 | Success Audit event |
| 4624    | 06-06-2026 13:05:56 | Success Audit event |
| 4624    | 06-06-2026 13:11:59 | Success Audit event |
| 4624    | 06-06-2026 13:19:53 | Success Audit event |
| 4624    | 06-06-2026 13:21:46 | Success Audit event |
| 4624    | 06-06-2026 13:28:33 | Success Audit event |
| 4624    | 06-06-2026 13:29:57 | Success Audit event |

SQL Elapsed: 00:00:01 Rows: 480 Log Type: EVTLOG

```
SELECT eventid,timegenerated,eventtypename
FROM '[LOGFILEPATH]'
WHERE EventID=4624
```

## **Observation**

The query returned multiple successful authentication events recorded within the Security log.

Each record contained:

- Timestamp
- Event ID
- Event Type

## **Analysis**

Successful logon events provide visibility into legitimate user activity and account usage patterns.

These events are often used during investigations to:

- Build user timelines
- Verify account activity
- Investigate unauthorized access
- Identify unusual login patterns

## **Key Finding**

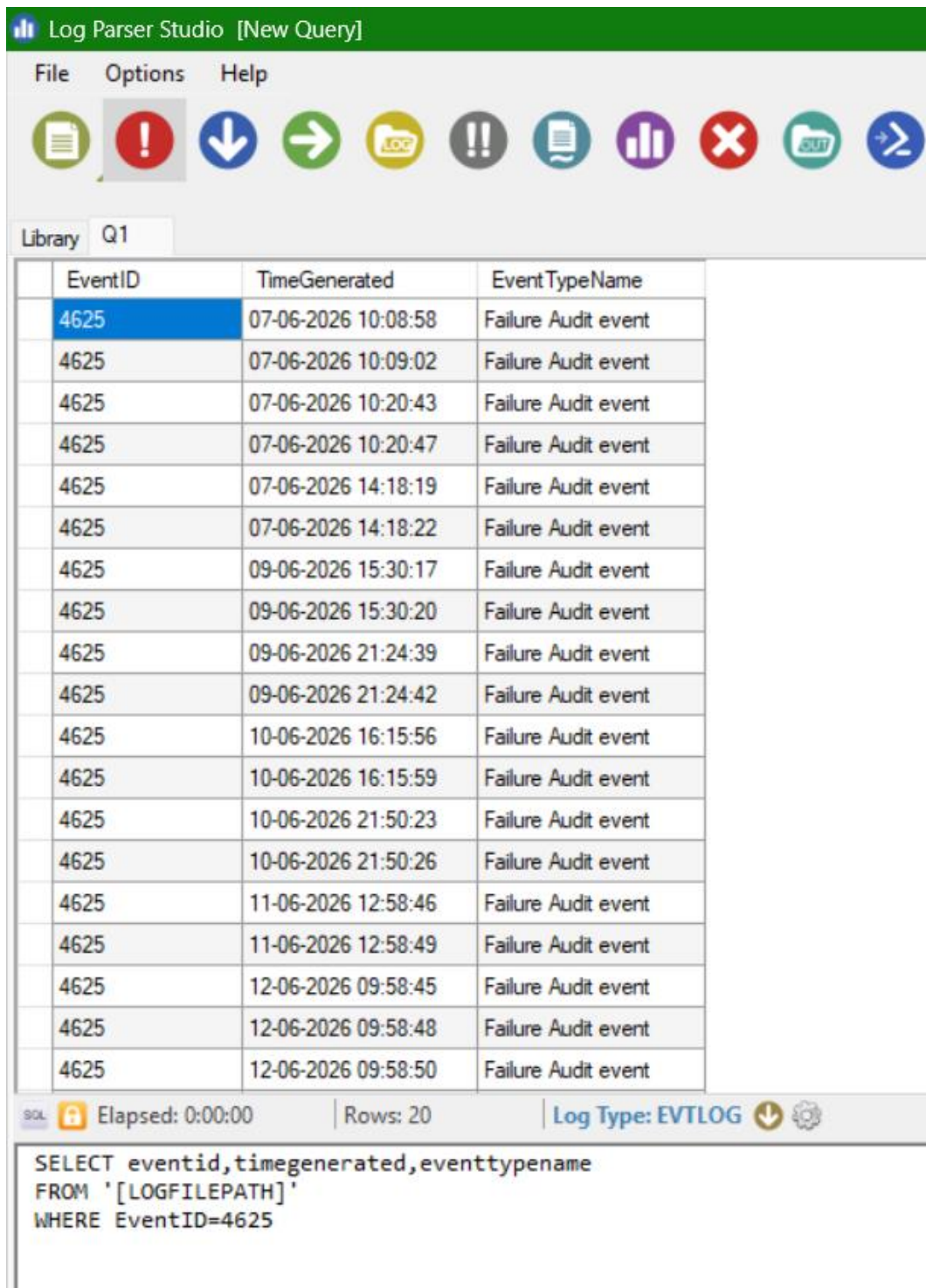
Event ID 4624 represented normal authentication activity observed during system usage.

## 6. Failed Authentication Analysis (Event ID 4625)

Event ID 4625 records failed authentication attempts.

### Query

```
SELECT EventID, TimeGenerated, EventTypeName
FROM '[LOGFILEPATH]'
WHERE EventID = 4625
```



The screenshot shows the Log Parser Studio interface. The title bar reads "Log Parser Studio [New Query]". The menu bar includes "File", "Options", and "Help". The toolbar contains icons for file operations, a warning icon, navigation, and execution. The "Library" tab is active, showing a table with 20 rows of data. The first row is highlighted in blue. The status bar at the bottom indicates "Elapsed: 0:00:00", "Rows: 20", and "Log Type: EVTLOG". The SQL query is displayed in the bottom pane.

| EventID | TimeGenerated       | EventTypeName       |
|---------|---------------------|---------------------|
| 4625    | 07-06-2026 10:08:58 | Failure Audit event |
| 4625    | 07-06-2026 10:09:02 | Failure Audit event |
| 4625    | 07-06-2026 10:20:43 | Failure Audit event |
| 4625    | 07-06-2026 10:20:47 | Failure Audit event |
| 4625    | 07-06-2026 14:18:19 | Failure Audit event |
| 4625    | 07-06-2026 14:18:22 | Failure Audit event |
| 4625    | 09-06-2026 15:30:17 | Failure Audit event |
| 4625    | 09-06-2026 15:30:20 | Failure Audit event |
| 4625    | 09-06-2026 21:24:39 | Failure Audit event |
| 4625    | 09-06-2026 21:24:42 | Failure Audit event |
| 4625    | 10-06-2026 16:15:56 | Failure Audit event |
| 4625    | 10-06-2026 16:15:59 | Failure Audit event |
| 4625    | 10-06-2026 21:50:23 | Failure Audit event |
| 4625    | 10-06-2026 21:50:26 | Failure Audit event |
| 4625    | 11-06-2026 12:58:46 | Failure Audit event |
| 4625    | 11-06-2026 12:58:49 | Failure Audit event |
| 4625    | 12-06-2026 09:58:45 | Failure Audit event |
| 4625    | 12-06-2026 09:58:48 | Failure Audit event |
| 4625    | 12-06-2026 09:58:50 | Failure Audit event |

```
SELECT eventid,timegenerated,eventtypename
FROM '[LOGFILEPATH]'
WHERE EventID=4625
```

## **Observation**

The query identified multiple failed authentication attempts within the Security log.

These entries represented unsuccessful logon attempts generated by the operating system.

## **Analysis**

Failed logon events are frequently reviewed because they may indicate:

- Incorrect passwords
- Expired credentials
- Account lockouts
- Password spraying attempts
- Brute-force attacks

Reviewing failed logon frequency can help determine whether authentication failures are normal user behavior or potential malicious activity.

## **Key Finding**

The identified Event ID 4625 entries represented failed authentication events that could be further investigated using account and source information.

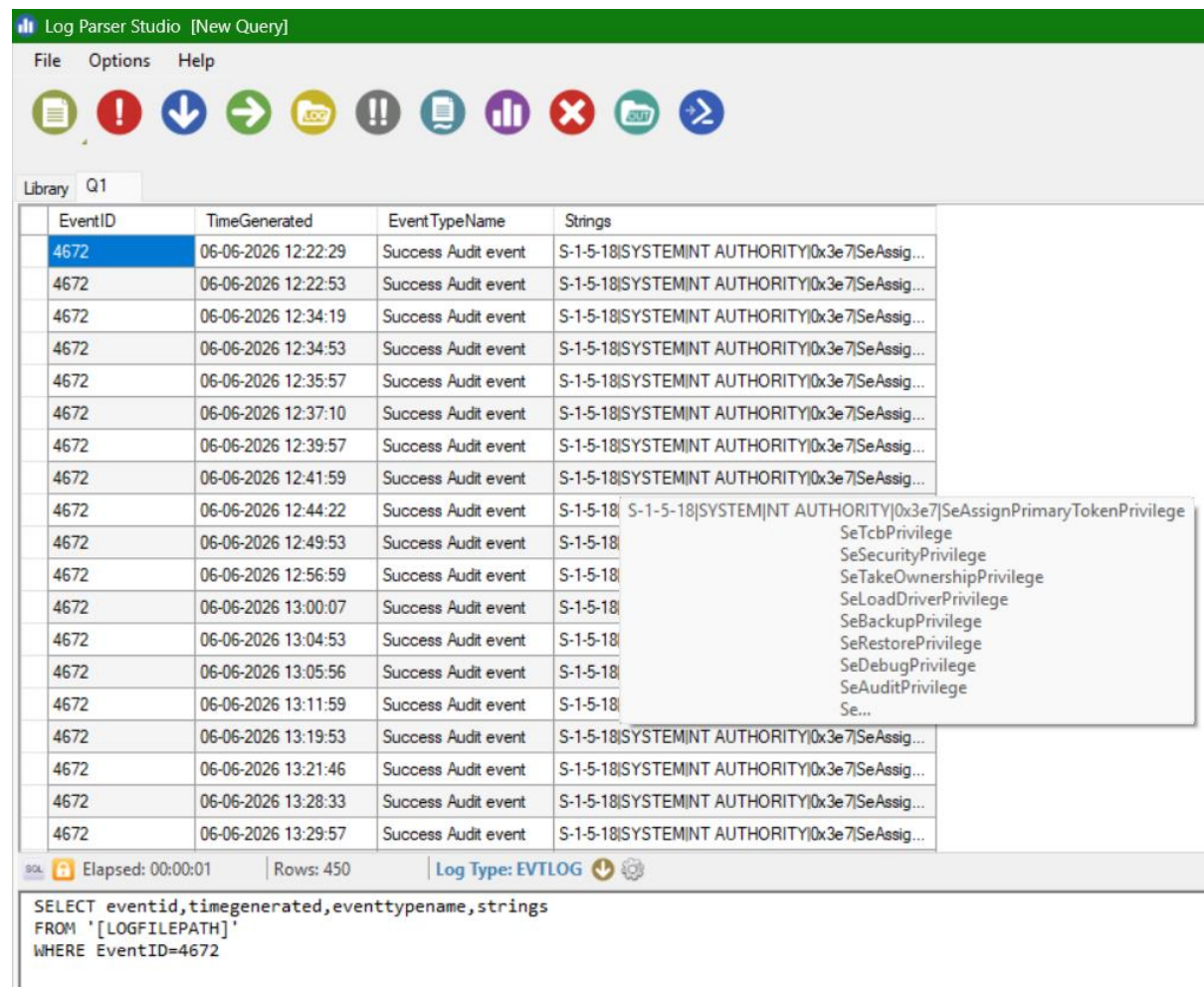


## 7. Event ID 4672 – Special Privileges Assigned to New Logon

Event ID 4672 records situations where elevated privileges are assigned to a user or service during authentication.

### Query Used

```
SELECT EventID, TimeGenerated, EventTypeName, Strings
FROM '[LOGFILEPATH]'
WHERE EventID = 4672
```



| EventID | TimeGenerated       | EventTypeName       | Strings                                                          |
|---------|---------------------|---------------------|------------------------------------------------------------------|
| 4672    | 06-06-2026 12:22:29 | Success Audit event | S-1-5-18 SYSTEM NT AUTHORITY 0x3e7 SeAssignPrimaryTokenPrivilege |
| 4672    | 06-06-2026 12:22:53 | Success Audit event | S-1-5-18 SYSTEM NT AUTHORITY 0x3e7 SeAssignPrimaryTokenPrivilege |
| 4672    | 06-06-2026 12:34:19 | Success Audit event | S-1-5-18 SYSTEM NT AUTHORITY 0x3e7 SeAssignPrimaryTokenPrivilege |
| 4672    | 06-06-2026 12:34:53 | Success Audit event | S-1-5-18 SYSTEM NT AUTHORITY 0x3e7 SeAssignPrimaryTokenPrivilege |
| 4672    | 06-06-2026 12:35:57 | Success Audit event | S-1-5-18 SYSTEM NT AUTHORITY 0x3e7 SeAssignPrimaryTokenPrivilege |
| 4672    | 06-06-2026 12:37:10 | Success Audit event | S-1-5-18 SYSTEM NT AUTHORITY 0x3e7 SeAssignPrimaryTokenPrivilege |
| 4672    | 06-06-2026 12:39:57 | Success Audit event | S-1-5-18 SYSTEM NT AUTHORITY 0x3e7 SeAssignPrimaryTokenPrivilege |
| 4672    | 06-06-2026 12:41:59 | Success Audit event | S-1-5-18 SYSTEM NT AUTHORITY 0x3e7 SeAssignPrimaryTokenPrivilege |
| 4672    | 06-06-2026 12:44:22 | Success Audit event | S-1-5-18 SYSTEM NT AUTHORITY 0x3e7 SeAssignPrimaryTokenPrivilege |
| 4672    | 06-06-2026 12:49:53 | Success Audit event | S-1-5-18 SYSTEM NT AUTHORITY 0x3e7 SeAssignPrimaryTokenPrivilege |
| 4672    | 06-06-2026 12:56:59 | Success Audit event | S-1-5-18 SYSTEM NT AUTHORITY 0x3e7 SeAssignPrimaryTokenPrivilege |
| 4672    | 06-06-2026 13:00:07 | Success Audit event | S-1-5-18 SYSTEM NT AUTHORITY 0x3e7 SeAssignPrimaryTokenPrivilege |
| 4672    | 06-06-2026 13:04:53 | Success Audit event | S-1-5-18 SYSTEM NT AUTHORITY 0x3e7 SeAssignPrimaryTokenPrivilege |
| 4672    | 06-06-2026 13:05:56 | Success Audit event | S-1-5-18 SYSTEM NT AUTHORITY 0x3e7 SeAssignPrimaryTokenPrivilege |
| 4672    | 06-06-2026 13:11:59 | Success Audit event | S-1-5-18 SYSTEM NT AUTHORITY 0x3e7 SeAssignPrimaryTokenPrivilege |
| 4672    | 06-06-2026 13:19:53 | Success Audit event | S-1-5-18 SYSTEM NT AUTHORITY 0x3e7 SeAssignPrimaryTokenPrivilege |
| 4672    | 06-06-2026 13:21:46 | Success Audit event | S-1-5-18 SYSTEM NT AUTHORITY 0x3e7 SeAssignPrimaryTokenPrivilege |
| 4672    | 06-06-2026 13:28:33 | Success Audit event | S-1-5-18 SYSTEM NT AUTHORITY 0x3e7 SeAssignPrimaryTokenPrivilege |
| 4672    | 06-06-2026 13:29:57 | Success Audit event | S-1-5-18 SYSTEM NT AUTHORITY 0x3e7 SeAssignPrimaryTokenPrivilege |

Log Parser Studio [New Query]

File Options Help

Library Q1

Elapsed: 00:00:01 Rows: 450 Log Type: EVTLOG

```
SELECT eventid,timegenerated,eventtypename,strings
FROM '[LOGFILEPATH]'
WHERE EventID=4672
```

### Observation

Multiple Event ID 4672 entries were identified during log analysis. The event data revealed several administrative privileges assigned during authentication, including:

- SeDebugPrivilege
- SeBackupPrivilege
- SeRestorePrivilege
- SeSecurityPrivilege
- SeTakeOwnershipPrivilege
- SeLoadDriverPrivilege

## **Analysis**

These privileges provide elevated access to sensitive operating system functions. Events of this type are commonly generated when administrative accounts, services, or SYSTEM-level processes authenticate.

Particular attention should be given to privileges such as **SeDebugPrivilege**, which allows interaction with other processes and is frequently associated with credential dumping and advanced post-exploitation techniques.

## **Key Finding**

Monitoring Event ID 4672 helps identify privileged account usage and provides visibility into sessions operating with elevated permissions.

Privileged logon events should always be correlated with successful authentication events to determine which accounts received elevated access.

Event ID 5379 records interactions with Windows Credential Manager.

```
SELECT EventID, TimeGenerated, EventType, Strings
FROM '[LOGFILEPATH]'
WHERE EventID = 5379
```

## Observation

A large number of Event ID 5379 records were identified during analysis. The event data showed applications and system processes accessing credentials stored within Windows Credential Manager.

Credential references associated with user accounts and application authentication activity were visible within the event data.

Credential Manager is used by Windows to securely store authentication information for websites, applications, network resources, and Microsoft services. Access to stored credentials generates Event ID 5379 entries.

While these events are often legitimate, they are highly relevant during investigations involving credential theft, credential harvesting, or lateral movement activities.

Frequent monitoring of Event ID 5379 can help identify unusual credential retrieval patterns and unauthorized access to stored authentication information.

Credential-related events provide valuable insight into how authentication data is accessed and used across the operating system.

## **9. Findings**

The investigation produced the following findings:

1. Windows Security Logs were successfully imported into Log Parser Studio.
2. Query-based analysis simplified the review of large event datasets.
3. Event frequency analysis identified the most common security events within the environment.
4. Event ID 4624 provided visibility into successful authentication activity.
5. Event ID 4625 highlighted failed authentication attempts.
6. Event ID 5379 generated the highest number of recorded events.
7. Log Parser Studio significantly improved log analysis efficiency compared to manual Event Viewer review.

## **10. Conclusion**

This investigation demonstrated how Log Parser Studio can be used to efficiently analyze Windows Security Logs through SQL-style queries. By importing EVTX files and executing targeted searches, authentication activity and event distributions were rapidly identified without manually reviewing individual log entries. The exercise highlighted the value of query-based log analysis for SOC operations, threat hunting, and incident response workflows.